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PLL design/analysis worksheet Version 3.8 (3/13/2002)

by **Lance Lascari** lance@RFdude.com

Based heavily on the national application note 1001 by William O. Keese entitled
"An Analysis and Performance Evaluation of Passive Filter Design Technique for Charge Pump Phase-Locked Loops"

Other References/sources of information:

A great deal of stuff on the transient analysis was drawn from the comprehensive text PLL Performance, Simulation, and Design By Dean Banerjee. The text is available free at http://www.national.com/appinfo/wireless/deansbook.pdf

Frequency Synthesizer Design Handbook by James A. Crawford (residual FM and RMS phase error, Zero Order Hold)

"Design of Wideband Frequency Synthesizers" by Jeff Blake **RF design Magazine**, May 1988 -sampling phase det .

"Source and Synthesizer Phase Noise Requirements for QAM Radio Applications", William Reuter, CTI

(available at the CTI website and at the microwave product digest website, from January 1999 issue of MPD)

Features:

1. charge-pump based PLL loop filter design and loop analysis (based on national app-note) + phase margin correction by me (using Blake's transfer function for the sampling phase detector).
2. Phase noise analysis by me, with lots of input from the references
3. RMS phase error calc, as well as residual FM calc from Crawford's book and my alterations here
4. Estimated adjacent channel rejection due to reciprocal mixing (integrated noise over the adj. channel) -me
5. integrated noise over specified bandwidth to estimate CNR/SNR for various communications systems -me
6. attempt at analyzing "two-point" modulation, I.E. FM modulation using a PLL (mod. ref. and VCO) -me
7. Transient analysis (Dean Banerjee's PLL book)
8. **Analysis of the phase noise contribution due to thermal noise generated in R2+R3 and op-amp noise -me**
9. Analysis of the effects of multiple cascaded synthesizers using data files (two now, easily expandable)-me
10. Analysis of the effects of an ideal multiplier/divider at the synthesizer output -me (evolving/fixing 5/17/2k).

Notes:

- **Nothing is guaranteed, use at your own risk, please let me know if you like this and what you think!!**

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LL 3/14/1999 [snip snip snip {notes got too long}]

For the multiplier/divider analysis here are notes on what's affected

1) desired output frequency is divided by mult to get the synthesizer operating frequency

2) BW_desired is divided by mult for the design equations, so the frequency domain noise profile of the output signal after multiply or divide will look like it has this bandwidth.

3) The individual phase noise contributors (VCO, TCXO, phase detector, etc.) will all be calculated and plotted for the actual PLL frequency (i.e. where the VCO is running). The total noise however is multiplied by mult (magnitude) and the offset frequency is divided by mult (so that the frequency axis expands or contracts).

4) the File I/O stuff for looking at the effect of multiple synthesizers is not affected, since the mult/div. effects are calculated before the files are written.

5) The mult/div stuff is more thrown in for design purposes. If you really are evaluating a synthesizer that's part of a multiplier or divider (and trying to optimize it or get lab results to match analysis), you're probably better off leaving mult=1 and just analyzing the synthesizer as-is.

11/2/1999 LL -playing with better integrated noise estimates for QAM/QPSK systems assuming 2nd order CR loop. Note -- integrated noise stuff now uses 2nd order CR loop, setup with the BW and damping coeff

1/2/2000 LL -added op-amp stuff, including op-amp noise and revised resistor noise accordingly

1/13/2000 LL -fixed a bug in total noise calculation w/amp.

1/17/2000 LL - plots now use log scale,faster, 1/20/2000 LL -reformatted graphs, added 2 synth noise ex

5/17/2000 LL -fixing multiplier effects on output phase noise curve

10/9/2000 LL -fixed refs, fixed zero order hold compensation/response function based on input by Alberto Villa of Italy. Thanks Alberto!

11/9/2000 LL -fixed 3rd pole calc. Found by Rich Place, seems to have been in error since nat. AN1001!

2/(25-28)/2001 LL -putting in some additional file-write capabilities, totally redesigned the multi-synth plan based on some interpolation techniques I saw someone do (don't know the source, otherwise I would credit it). Mucho-faster for those portions! I also added a very cool SNR vs. upper integration limit plot

3/1/2001 LL - set TOL manually to get better accuracy on tolerance of integrals, now my SNR plot (newly added) behaves as it should. **3/20/2001** -normalized closed loop gain (hooks in several places).

1/1/2002 LL - stripped out combined noise analysis (and some other stuff) and put in separate worksheet, fixed some things that Mark Smith found and passed along relating to the resistor/op-amp noise analysis in the active case (thanks Mark!) I used his findings and reworked the analysis in a way I was most comfortable with

3/13/2002 LL - fixed two things 1) mathcad2001i barfs on anything with units being written to a file, 2) Correction by Dick Frey noting the error in calculations of filter transfer function when amplifier is used (I've never used the amplifier and may strip it out someday, but this was in fact an error that needed to be corrected -- thanks for the error Dick!)

System inputs (I'll try and keep these in the left-most column, and in blue)

important for integration

$f_{out_desired} := 445 \cdot MHz$ desired output frequency (post divider/multiplier)

$TOL := 0.000001$

$mult := 1$ multiplication/division ratio of VCO ouptut.
SET MULT to 2 for a *2 multiply at synth output, 0.5 for a /2

$f_{out} := f_{out_desired} \cdot \frac{1}{mult}$ $f_{out} = 445\,MHz$ VCO output freq

$f_{ref} := 100 \cdot kHz$ reference (phase detector) frequency

$K_{vco} := 20 \frac{MHz}{volt}$ VCO gain constant

$dBc := 1$

$msec := 1 \cdot 10^{-3} \cdot sec$

amplifier in the loop (for both noise and loop transfer function analysis)
!! make loop_amp_enabled equal to zero and set amp_gain=1 if you're not using one !!

$loop_amp_enabled := 0$ enables or disables loop amplifier

$amp_gain := 1$ sets amplifier gain (should only come into effect if amp is enabled)

$v_{amp_input_noise} := 4.5 \cdot 10^{-9} \cdot volt$ units are really volts/sqrt(Hz)
For the "*low noise*" LM833 from national, this is about 4.5 nV

This next part allows you to vary the charge pump current. Some synthesizers have this facility, if you want to simplify things, just set the current directly. Otherwise set the "base" number and the "step" number.

$K_{\phi_step} := 16$ index of charge pump current (change this to modify fraction of full scale current)

$K_{\phi_base} := 1 \cdot mA$ Max charge pump current

$K_{\phi_steps} := 16$ number of discrete steps in charge pump current

$K_{\phi} := K_{\phi_base} \cdot \frac{K_{\phi_step}}{K_{\phi_steps}}$ Phase Detector gain constant

$K_{\phi} = 1\,mA$ Actual Charge pump output

$BW_{desired} := 2 \cdot kHz$ **Desired loop bandwidth (post multiplier/divider)**

$BW := BW_{desired}$ $BW = 2 \times 10^{3}\,Hz$

$\phi p_{desired} := 55 \cdot deg$ desired phase margin

Now compensate for the phase shift (due to delay) in the phase detector when selecting the phase margin to compute
from... (only significant when the loop bandwidth approaches a significant fraction of the reference frequency)

$$G_{pd}(f) := \frac{1 - \exp\left(-2 \cdot \pi \cdot f \cdot i \cdot \frac{1}{f_{ref}}\right)}{\frac{2 \cdot \pi \cdot f \cdot i}{f_{ref}}}$$

Phase detector term from zero order hold
(significant as the loop bandwidth gets wide relative to the reference frequency. From Blake paper and Crawford reference) - this will be used later too. This was in error, and was fixed 10/9/2000 based on input by Alberto Villa. Thanks!

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$$\phi_{pd}(f) := \arg\left(G_{pd}(f)\right)\cdot\frac{360\cdot deg}{2\cdot\pi} \qquad \phi_p := \phi_{p_desired} - \phi_{pd}(BW)$$

$\phi_p = 58.6\,deg$

This is the corrected phase margin for the other computations (the delay in the phase detector is subtracted to precompensate the "requested" phase margin from the loop filter)

$ATTEN := .01$

Desired Attenuation in dB for reference spurs (be careful with this, think about it the added R3-C3 combination can only roll off at 6 dB per octave, so be very aware of the ratio of your reference frequency and your desired loop bandwidth, how many octaves away?)

Noise related parameters

$f_{tcxo} := 12.8\cdot MHz$ TCXO/XTAL oscillator output frequency

$N_{tcxo_ref} := -135\cdot dBc$ TCXO noise (vendor X on web shows -135 on 12.8 MHz tcxo)

$f_{tcxo_ref} := 100\cdot Hz$ offset frequency where TCXO noise is specified

$N_{vco_ref} := -115dBc$ VCO noise

$f_{vco_ref} := 10\cdot kHz$ offset frequency where VCO noise is specified

$N_{pd_ref} := -201\cdot dBc$ Phase noise of phase detector specified in a 1 Hz BW. Equations below account for actual reference frequency
For LMX235x, RF side is -201, IF side is -211
depends on VCC and on whether IF side is powered up or not

integrated noise setup (for linear modulation schemes, otherwise noise is analyzed over whole range set below) (SNR and RMS phase error estimates integrate over this portion of the curve)

$f_{cr_bw} := 0.2\cdot kHz$ Carrier recovery loop BW for QAM systems

$\zeta_{cr} := 0.7071$ Damping coeff. for 2nd order carrier rec.

adjacent channel stuff (for estimating adjacent channel rejection due to reciprocal mixing)
-this defines the segment of the phase noise curve that is integrated to produce a relative power that would fold back on channel

the curve is integrated across a segment of adj. ch. bw centered at the adjacent channel spacing

$f_{channel_spacing} := 25\cdot kHz$ center to center channel spacing

$f_{adj_ch_bw} := 20\cdot kHz$ defines bandwidth over which noise should be integrated

$$f_{adj_channel_int_edge} := \frac{f_{channel_spacing}}{2} + \frac{f_{channel_spacing} - f_{adj_ch_bw}}{2}$$

$f_{adj_channel_int_edge} = 15\,kHz$ defines nearest edge of adjacent channel (above or below)

$Rejection_{co_channel} := -12$ what relative level of noise (dBc) does it take to degrade SNR/sensitivity (i.e. depends on modulation type of course)

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Modulation stuff

-For an FM/FSK system the peak deviation can be entered to calculate the SNR once the residual FM is calculated

$deviation_{pk} := 3 \cdot kHz$

Analysis setup.... Frequency for analysis (involving integration, etc)

$f_{start} := 10 \cdot Hz$ $f_{stop} := 10 \cdot MHz$

Here's the new setup for much of the sweeping (used in file I/O). I recommend using at least 20 points in this new log sweep scheme

$swp_points := 20$ $spts := 0, 1 .. swp_points$

actually the real number of points is swp_points+1

$$swp_{frq_{spts}} := f_{start} \cdot 10^{\left(\frac{spts}{swp_points} \cdot \log\left(\frac{f_{stop}}{f_{start}}\right)\right)}$$

$$swp_{frq_{swp_points}} = 10 \times 10^{6}\ time^{-1}$$

plotting setup.... Frequency for plotting, etc. (Noise and some graphs plotted over this range, log=fast)

$f_{plotstart} := 10 \cdot Hz$ $f_{plotstop} := 10 \cdot MHz$ $plotpoints := 100$ $ppts := 0, 1 .. plotpoints$

$$pltfreqs_{ppts} := f_{plotstart} \cdot 10^{\left(\frac{ppts}{plotpoints} \cdot \log\left(\frac{f_{plotstop}}{f_{plotstart}}\right)\right)}$$

END of main user input area

Loop cutoff frequency/ Phase inflection point

$$\omega_p := \frac{2 \cdot \pi}{rad} \cdot BW \qquad \omega_p = 1.25664 \times 10^{4}\ \frac{rad}{sec}$$

$$\tau_1 := \frac{\sec(\phi_p) - \tan(\phi_p)}{\omega_p} \qquad \tau_1 = 2.23682 \times 10^{-5}\ sec$$

Additional attenuation due to the extra pole (to be added)

$$\tau_3 := \sqrt{\frac{10^{\left(\frac{ATTEN}{10}\right)} - 1}{\left(2 \cdot \pi \cdot f_{ref}\right)^2}} \qquad \tau_3 = 7.64149 \times 10^{-8}\ sec$$

$$\omega_c := \frac{\tan(\phi_p) \cdot (\tau_1 + \tau_3)}{\left[(\tau_1 + \tau_3)^2 + \tau_1 \cdot \tau_3\right]} \cdot \left[\left[\sqrt{\frac{(\tau_1 + \tau_3)^2 + (\tau_1 \cdot \tau_3)}{\left[\tan(\phi_p) \cdot (\tau_1 + \tau_3)\right]^2} + 1}\right] - 1\right] \qquad \omega_c = 1.25205 \times 10^{4}\ time^{-1}$$

this is now the 2nd time constant adjusted for the third pole's presense...

$$\tau_2 := \frac{1}{\omega_c^{\,2} \cdot (\tau_1 + \tau_3)} \qquad \tau_2 = 2.84214 \times 10^{-4}\ sec$$

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$$N := \frac{f_{out}}{f_{ref}} \qquad R := \frac{f_{tcxo}}{f_{ref}}$$

Calculate N and R

C1 is also adjusted based on the time constant of the third pole

$$C_1 := \frac{(K_\phi \cdot K_{vco} \cdot \tau_1)}{\omega_c^{\,2} \cdot N \cdot \tau_2} \cdot \sqrt{\frac{\left(1 + \omega_c^{\,2} \cdot \tau_2^{\,2}\right)}{\left(1 + \omega_c^{\,2} \cdot \tau_1^{\,2}\right)\left(1 + \omega_c^{\,2} \cdot \tau_3^{\,2}\right)}} \qquad C_1 = 8.03135 \times 10^{-3}\,\mu F$$

$$C_1 := \begin{cases} (C_1 \cdot \text{amp_gain}) & \text{if loop_amp_enabled} = 1 \\ C_1 & \text{otherwise} \end{cases}$$

3/13/2002 -- fixed so that gain isn't included if amp isn't on

$$C_2 := C_1 \cdot \left(\frac{\tau_2}{\tau_1} - 1\right) \qquad C_2 = 0.09402\,\mu F$$

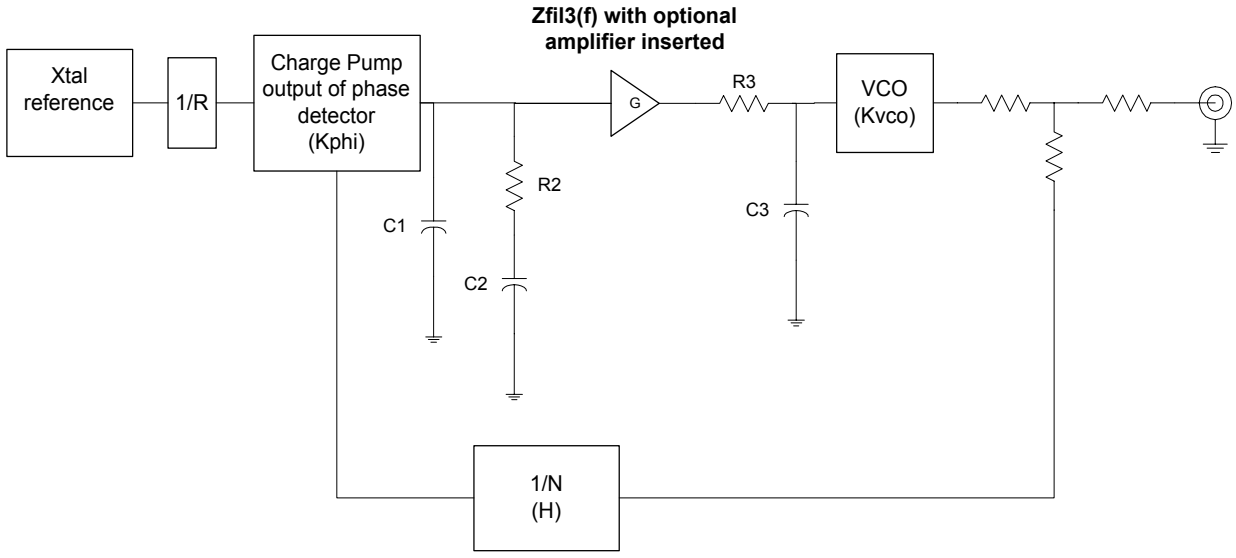
$$R_2 := \frac{\tau_2}{C_2} \qquad R_2 = 3.02303\,k\Omega$$

Use a rule of thumb for selecting R3 -- is it good enough? (keep it from loading filter, among other things)

$$R_3 := 2 \cdot R_2 \qquad R_3 = 6.04607\,k\Omega$$

$$C_3 := \frac{\tau_3}{R_3} \qquad C_3 = 12.63878\,pF$$

Basic PLL diagram



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Redefine component values here if desired (for realistic values)....

Computed values

$C_1 = 8.03135 \times 10^{-3}\,\mu F$ $C_2 = 0.09402\,\mu F$ $R_2 = 3.02303\,k\Omega$ $R_3 = 6.04607\,k\Omega$ $C_3 = 12.63878\,pF$

Plug in your real values here (or delete/rename these below to use the computed values) Remove underscore "_" to insert your values (below) and use for analysis.

$_C_1 := 0.1 \cdot \mu F$ $_C_2 := 1 \cdot \mu F$ $_R_2 := 0.5 \cdot k\Omega$ $_R_3 := 1 \cdot k\Omega$ $_C_3 := 4700 \cdot pF$

I would argue that one often wants to play with R3 & C3 rather than let the equations pick them, it's a dangerous portion of the circuit (that I usually leave out!). In any case, here's a quick calc of it's cutoff.

$$f_{3rdpole} := \frac{1}{2 \cdot \pi \cdot R_3 \cdot C_3} \qquad f_{3rdpole} = 2.08277 \times 10^{3}\,kHz$$

Loop filter impedances/transfer functions

$$Z_{fil}(f) := \frac{\left(1 + R_2 \cdot 2 \cdot \pi \cdot f \cdot i \cdot C_2\right)}{\left[2 \cdot \pi \cdot f \cdot i \cdot \left(C_2 + C_1 + R_2 \cdot 2 \cdot \pi \cdot f \cdot i \cdot C_1 \cdot C_2\right)\right]}$$

second order filter impedance equation

$$Z_{fil3_basic}(f) := \frac{Z_{fil}(f) \cdot \frac{1}{2 \cdot \pi \cdot f \cdot i \cdot C_3}}{Z_{fil}(f) + R_3 + \left(\frac{1}{2 \cdot \pi \cdot f \cdot i \cdot C_3}\right)}$$

third order filter transfer function when combined without an amplifier (all loading accounted for)

$$Z_{fil3_withamp_equiv}(f) := Z_{fil}(f) \cdot \left[\frac{\left(\frac{1}{2 \cdot \pi \cdot f \cdot i \cdot C_3}\right)}{\left[R_3 + \left(\frac{1}{2 \cdot \pi \cdot f \cdot i \cdot C_3}\right)\right]}\right] \cdot amp_gain$$

third order transfer function, with amplifier (in the sense that the 2nd order filter is isolated and not loaded by the 3rd pole), i.e. it's the cascade of two non-interacting transfer functions 3/13/2002

$$Z_{fil3}(f) := \begin{cases} Z_{fil3_withamp_equiv}(f) & \text{if } loop_amp_enabled = 1 \\ Z_{fil3_basic}(f) & \text{otherwise} \end{cases}$$

conditionally evaluate the transfer function depending on whether an amplifier is in the loop or not. (previous error pointed out by R. L. Frey 3/12/2002)

these "definitions" from the national app-note.

$$G(f) := \frac{K_\phi \cdot Z_{fil3}(f) \cdot K_{vco}}{2 \cdot \pi \cdot f \cdot i} \cdot G_{pd}(f)$$

Forward loop gain

$$H := \frac{1}{N}$$

Reverse loop gain

$$G_{ol}(f) := G(f) \cdot H$$

open loop gain

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$$G_{cl}(f) := \frac{G(f)}{1 + G(f)\cdot H}\cdot H$$

closed loop gain **(now normalized)**

Find actual bandwidth and calculate phase margin there

$x := 1\cdot Hz$

$$BW_{actual} := root\left(20\cdot\log\left(\left|G_{ol}(x)\right|\right), x\right)$$

Actually look at the open loop response to find the actual loop bandwidth (where G=0dB)

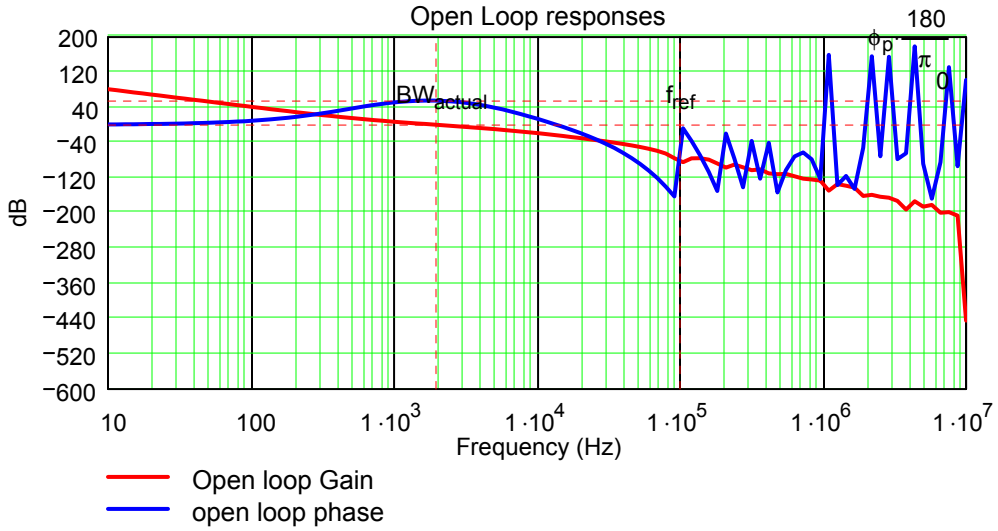
$$phase_{ol}(f) := \left(arg\left(-G_{ol}(f)\right)\cdot\frac{180}{\pi}\right)$$

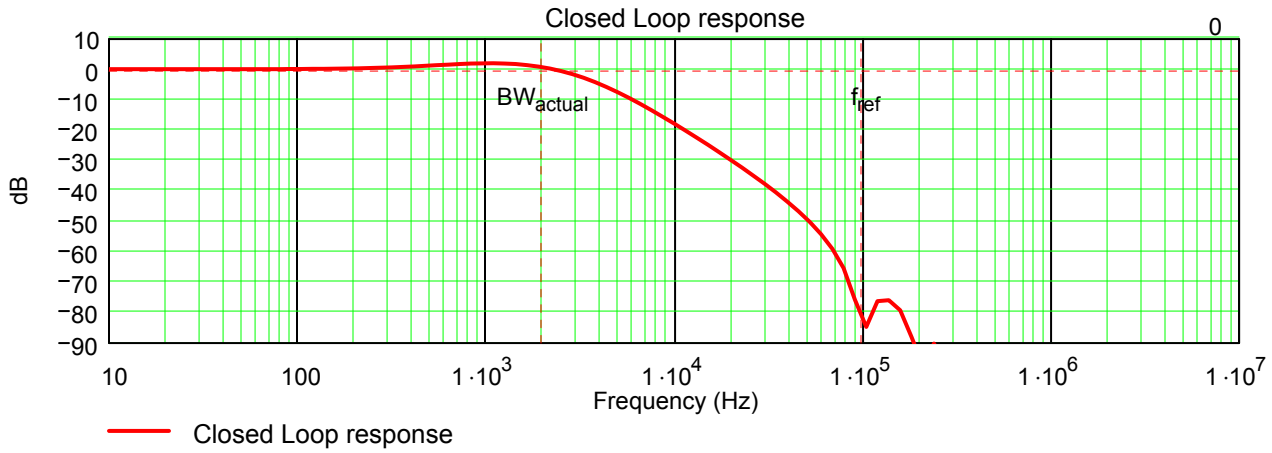
for open loop analysis of phase margin (reformatted 1/1/2001 -- better display usage, thanks Mark.)

Evaluate the phase margin at the actual loop bandwidth

$BW_{actual} = 1.99116\,kHz$

$phase_{ol}(BW_{actual}) = 54.99778$





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Phase noise --- the fun stuff.

General purpose equation for phase noise of an oscillator (applicable to tcxo and vco)

note: different noise sources will be shaped by different loop transfer functions

fofs = offset for calc

fref= reference frequency where noise is specified in dBc/Hz

Pref = power level specified at fref (dBc)

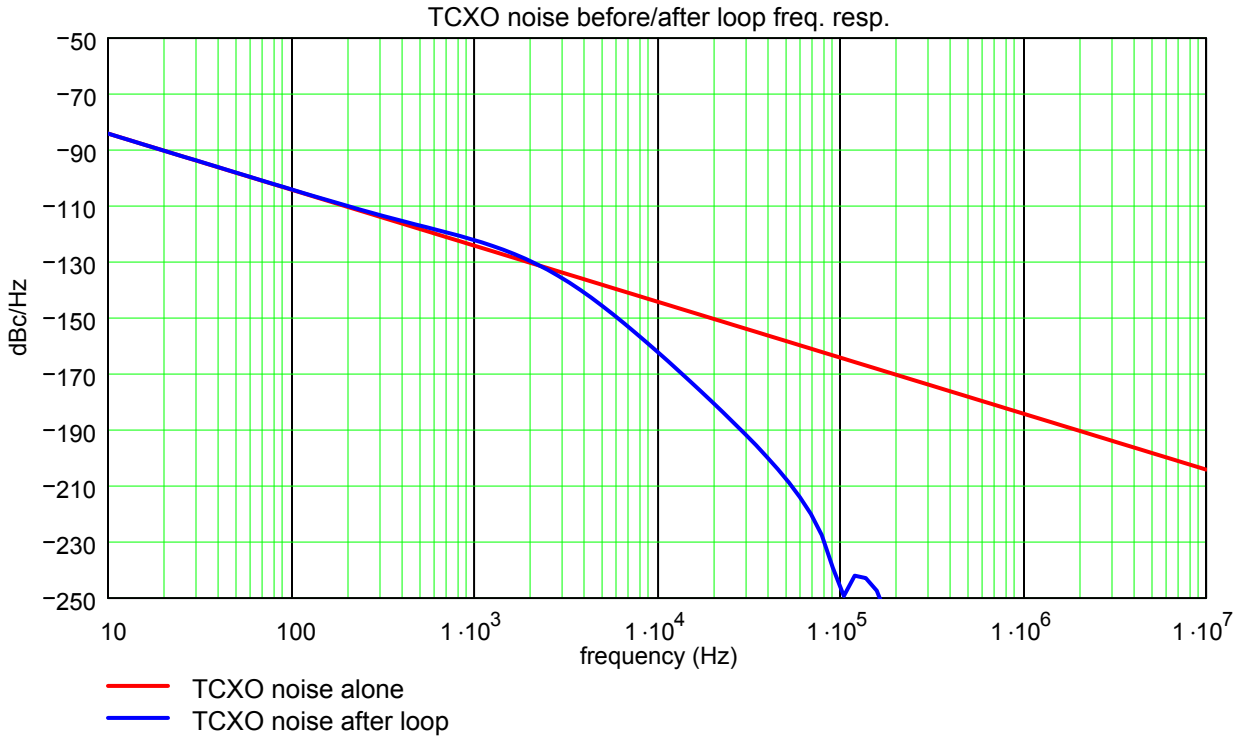
output of equation is the equivalent noise voltage

$$\text{Noise}(f_{ofs}, f_{ref}, P_{ref}) := \frac{10^{\left(\frac{P_{ref}}{20}\right)}}{\frac{f_{ofs}}{f_{ref}}}$$

TCXO Noise

The 1/R term is required to get the TCXO noise contribution accurately. The Loop response accounts For N, but the phase noise contribution of the TCXO is proportional to R and N, rather than Just N If N weren't part of the loop response then we would multiply by fout/ftcxo

$$N_{tcxo}(f) := \text{Noise}(f, f_{tcxo_ref}, N_{tcxo_ref}) \cdot \left(G_{cl}(f) \cdot \frac{N}{R}\right)$$

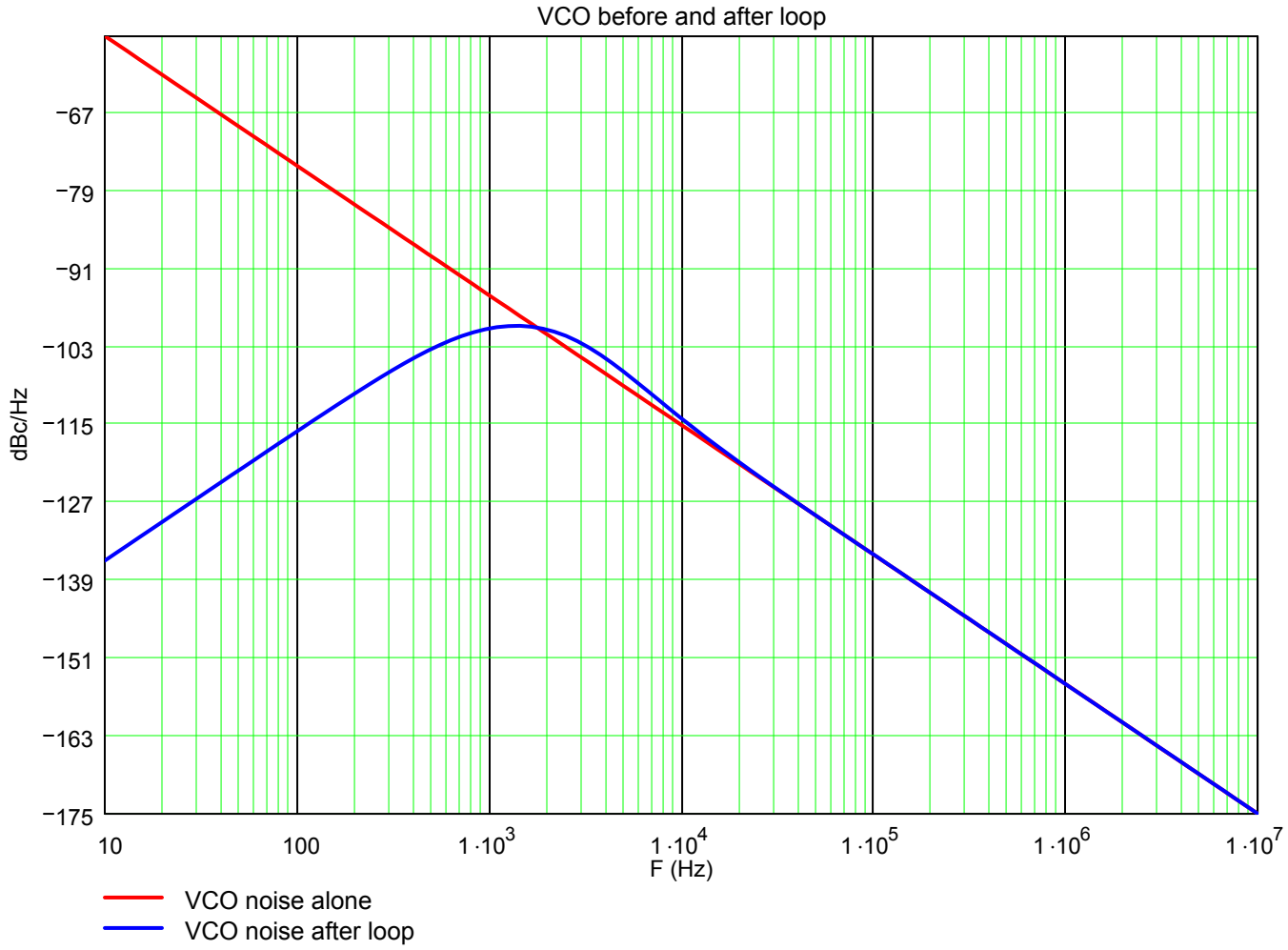


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If it weren't for the phase detector noise, etc.. Life would be grand... the VCO noise is greatly attenuated in the loop.

$$N_{vco_alone}(f) := \text{Noise}\left(f, f_{vco_ref}, N_{vco_ref}\right)$$

$$N_{vco}(f) := \text{Noise}\left(f, f_{vco_ref}, N_{vco_ref}\right) \cdot \frac{1}{1 + G_{ol}(f)}$$



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Complete analysis of resistor & loop amplifier (OP-AMP) noise contributing to overall Phase noise in the passive charge pump PLL synthesizer.

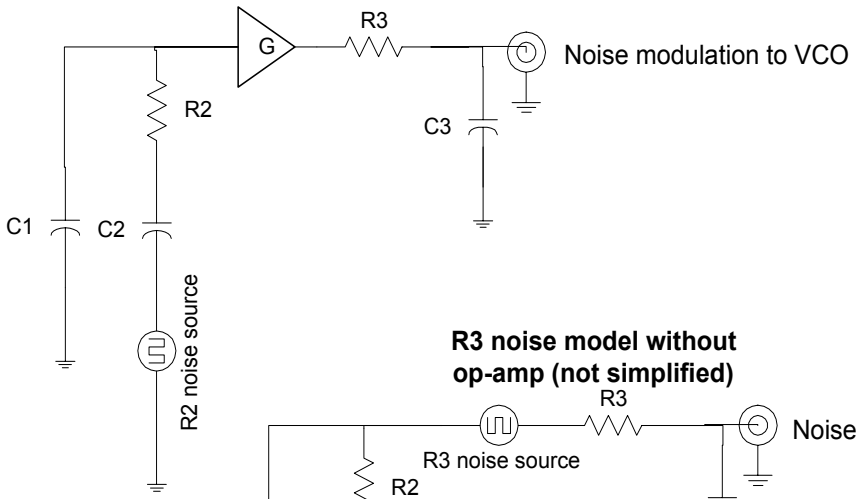
These basic analysis techniques can be easily applied to any general resistor noise problem (or op-amp noise problem) , i.e. different loop topologies (if you're good at algebra).First pass equations are not simplified all that much, I ran into errors and left them as-is. This treats the problem as "outside of the loop" at first.

Note: 1) each noise source is assumed to be uncorrelated.

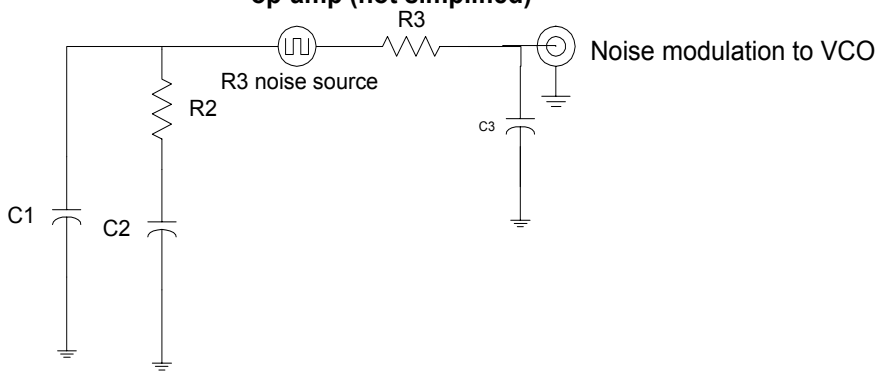
2) Resistor noise is independent of DC current/voltage.

Important!!! set Rnoise_enabled to zero if you don't want to include this in all of the later noise analysis!

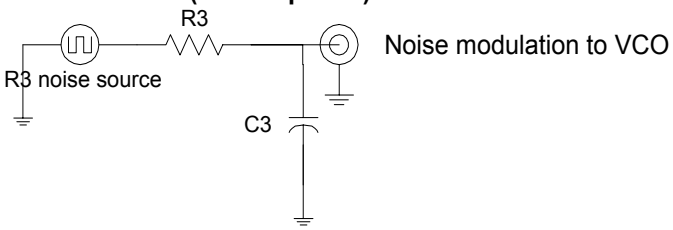
R2 Noise mode with or without op-amp (not simplified)



R3 noise model without op-amp (not simplified)



R3 noise model with op-amp -- assume op-amp has very low output impedance (not simplified)



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Rnoise_enabled := 1 Important -- set Rnoise_enabled --

$k := 1.38 \cdot 10^{-23} \left(\frac{joule}{K} \right)$ Boltzmann's Constant

$Temp := 300 \cdot K$ Temperature in Kelvins

$V_{resistor_noise}(R) := \sqrt{4 \cdot k \cdot Temp \cdot R \cdot Hz}$ RMS noise voltage in volts

$V_{n_R3} := V_{resistor_noise}(R_3)$ $V_{n_R2} := V_{resistor_noise}(R_2)$

$V_{n_R3} = 1.00061 \times 10^{-8}\, volt$ $V_{n_R2} = 7.07541 \times 10^{-9}\, volt$

Baseband noise models (pre VCO-modulation)

To simplify the circuit analysis, here are some smaller equations for the noise sources. In the case of R2, ~~circuit is broken down into a thevenin equivalent network with the voltage source in series with R2, C2. The output voltage is derived from a voltage divider composed shunt cap C3, and all of the rest of the network reduced to an impedance in series with the noise voltage source.~~ As of 1/1/2002, I redid some of this analysis per the suggestion of a kind observer. At this point the noise voltage due to R2 is derived at the "top" of the conventional C1, R2, C2 loop filter. For the passive case (no amplifier), this voltage is the voltage derived from a voltage source in series with R2 and C2, with the "divider" load being C1 in parallel with the impedance due to R3+C3. For the active case (ideal high input Z op-amp), the noise at this node is not loaded by R3 and C3. For both cases, the resulting voltage at that node is multiplied by the R3/C3 divider (with the amplifier there is merely gain inserted as well). Previously I was treating the R2 noise identically for both cases which was an error (but depending on the case it may have been a fairly minor error, who knows really).

At this point I've broken the equations into two sets, one for when the op-amp is enabled, one for when it is not.

Important assumptions about the effect of the op-amp in case #2

- As shown, the op-amp is inserted between the 2nd order loop filter and the third pole, so that the 3rd pole may roll off the op-amp noise.
- R2 noise gets amplified by the gain of the op-amp, then filtered by R3, C3
- Op-amp noise is the equivalent input noise voltage, amplified by the op-amp gain, and then is filtered by R3, C3
- I will assume that the op-amp has a low output impedance, hence the R3 analysis is just that of R3 grounded at one end, then filtered by R3, C3. The gain of the op-amp does not affect the R3 noise.

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Case #1/ #2 with or without op-amp
R2 noise

Define the following Impedance equations

$$Z_{R2_1}(f) := R_2 + \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_2}$$ R2 series with C2

$$Z_{R2_2}(f) := \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_1}$$ C1 impedance

$$Z_{R2_3}(f) := R_3$$

$$Z_{R2_4}(f) := \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_3}$$ C3 impedance

$$Z_{R2_5}(f) := \frac{Z_{R2_2}(f)\cdot\left(Z_{R2_3}(f) + Z_{R2_4}(f)\right)}{Z_{R2_2}(f) + \left(Z_{R2_3}(f) + Z_{R2_4}(f)\right)}$$ Z1 in parallel with Z2 and R3+C3

This revised analysis was prompted by Mark Smith, as he found some errors. The error was as follows
For the amp-included case, the R2 noise was being calculated improperly as there is no effective loading of the loop filter by R3/C3. I ended up reworking the analysis anyway, in a slightly different form (it's been two years since I looked at this portion of the sheet, so I merely approached the circuit reduction differently)

$$\mu V := 10^{-6}\cdot volt$$

$$V_{noise_R2_withoutamp}(f) := V_{n_R2}\cdot\frac{Z_{R2_5}(f)}{Z_{R2_1}(f) + Z_{R2_5}(f)}\cdot\frac{Z_{R2_4}(f)}{Z_{R2_3}(f) + Z_{R2_4}(f)}$$

$$V_{noise_R2_withamp}(f) := V_{n_R2}\cdot amp_gain\cdot\frac{Z_{R2_2}(f)}{Z_{R2_1}(f) + Z_{R2_2}(f)}\cdot\frac{Z_{R2_4}(f)}{Z_{R2_3}(f) + Z_{R2_4}(f)}$$

$$V_{noise_R2}(f) := \begin{cases} V_{noise_R2_withamp}(f) & \text{if } loop_amp_enabled = 1 \\ V_{noise_R2_withoutamp}(f) & \text{if } loop_amp_enabled = 0 \end{cases}$$

Case #1, without op-amp
R3 noise

Define the following Impedance equations

Note, this one is easier since the voltage source is where it is (R3) connected between two impedances

$$Z_{R3_1}(f) := \frac{\frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_1}\cdot\left(R_2 + \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_2}\right)}{R_2 + \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_1} + \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_2}} + R_3$$

Parallel combination of C1 and C2 series with R2
Plus R3 in series

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$$Z_{R3_2}(f) := \frac{1}{2\cdot\pi\cdot f\cdot i\cdot C_3}$$ Impedance of C3

$$V_{noise_R3_withoutamp}(f) := V_{n_R3}\cdot\frac{Z_{R3_2}(f)}{Z_{R3_1}(f) + Z_{R3_2}(f)}$$ Baseband RMS noise due to R3

Case #2, with op-amp

R3 noise

Note, this one is much easier since we're assuming the op-amp has a very low output impedance (short to ground)

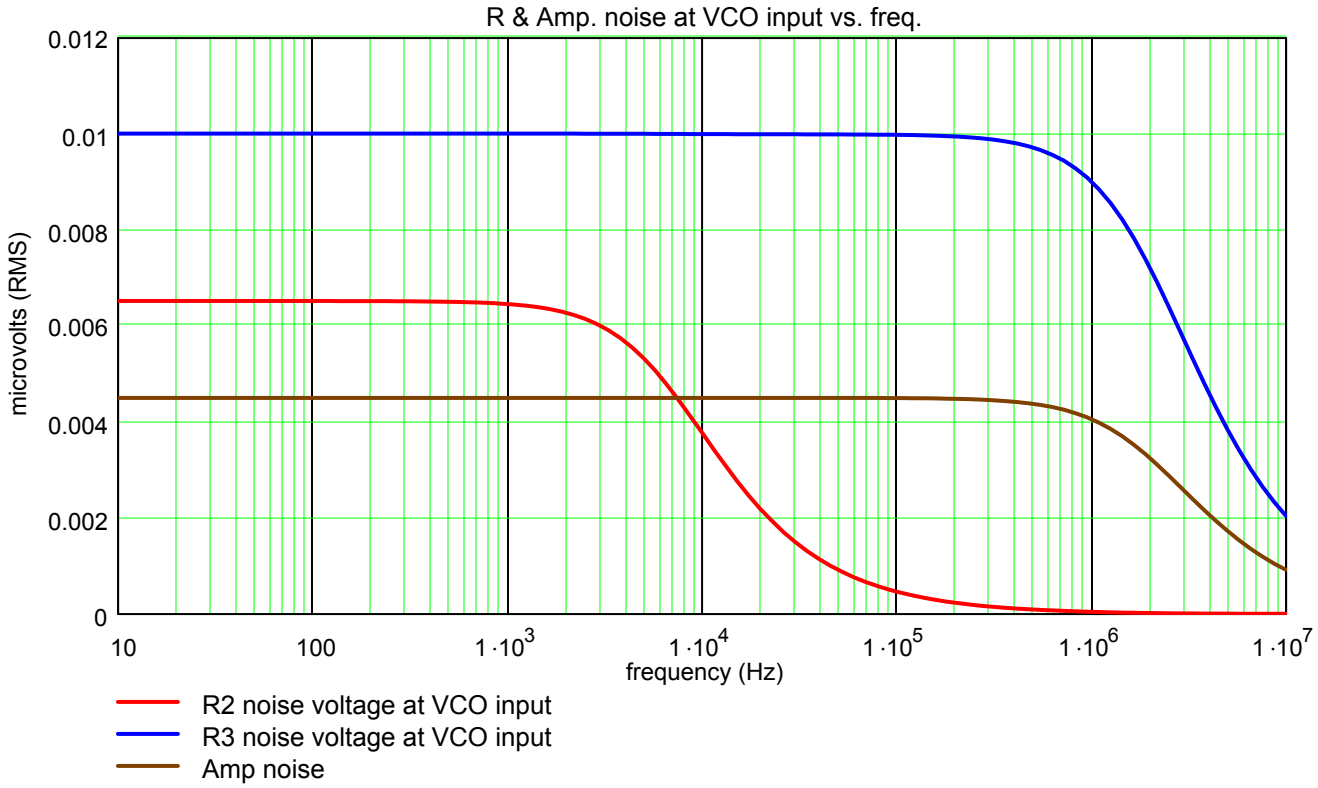
$$V_{noise_R3_withamp}(f) := V_{n_R3}\cdot\frac{Z_{R3_2}(f)}{R_3 + Z_{R3_2}(f)}$$ Baseband RMS noise due to R3

$$V_{noise_R3}(f) := \begin{cases} V_{noise_R3_withamp}(f) & \text{if } loop_amp_enabled = 1 \\ V_{noise_R3_withoutamp}(f) & \text{if } loop_amp_enabled = 0 \end{cases}$$

Amplifier in the loop's noise

-Op amp's input-referred noise voltage is amplified by the gain, then filtered by R3, C3

$$V_{noise_amp}(f) := amp_gain\cdot v_{amp_input_noise}\cdot\frac{Z_{R3_2}(f)}{R_3 + Z_{R3_2}(f)}$$ almost identical to above



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Now the equations for these sources modulating the VCO -- how we relate RMS noise voltage to frequency deviation, and how that maps into SSB phase noise.

--first a brief explanation of the FM modulation equations in symbolic form with an example (for my reference as much as anyone) . Thanks to Rich place for a good explanation of this, I'm sure I haven't captured it perfectly, but it's a shot.

-- note, the following examples hold for low modulation indexes (such as those we're targetting), where the modulation index is below 1. See any good communications text for references to very low modulation index systems, narrowband FM. Example: Communications Systems, A. Bruce Carlson in the third edition, this falls under chapter 7, Exponential CW Modulation.

specifically he describes the case where the modulation index <<1, the output signal (using the nomenclature that I'll use throughout) is represented by the following approximation:

$$f_{carrier_FM_modulated}(t) := \left[A_c \cdot \cos(f_c \cdot t) - \left[\frac{A_c \cdot m}{2} \cdot \cos\left[(f_c - f_m) \cdot t \right] \right] \right] + \frac{A_c \cdot m}{2} \cdot \cos\left[(f_c + f_m) \cdot t \right]$$

where A_c is the carrier amplitude in volts, f.c is the carrier frequency, f_m is the "tone" modulation frequency, and m is the modulation index (Beta in Carlson)

the time domain equation for an FM modulated carrier above can be represented (as is done in Carlson) by a phasor diagram with the main carrier phasor spinning at the carrier frequency with an amplitude of A_c, and two other phasors spinning at the message frequency rate (f_m) in opposite directions in quadrature at the tip of the carrier phasor. The amplitude of each of these two phasors is m*A_c/2.

First of all, the noise we will calculate from the resistors will be RMS noise voltage. This noise will then modulate A VCO. Using the approximation that at any given frequency, the noise will have a sinusoidal peak to average ratio (what else can we assume?), we need to multiply the RMS voltage by sqrt(2) to get the peak voltage.

This peak voltage will map directly to a peak deviation as the voltage will multiply the VCO gain (K_{vco}). The result of that multiplication is peak deviation. So the general approach will be as follows

1. Calculate RMS noise voltage coming from the resistors in the loop filter
2. Calculate peak deviation due to this voltage by obtaining the Peak noise voltage and multiplying that by the VCO gain constant, K_{vco}. This will be done at each frequency since the loop filter's noise is not flat
3. Calculate the modulation index for each frequency using the deviation and the frequency
4. Compute a curve of SSB phase noise directly from the modulation index .

This Example shows the FM modulation/deviation/modulating frequency relationships

$A_V := 1 \cdot 10^{-6} \cdot volt$ This is the modulating audio voltage, RMS at the modulating frequency f_m below

$$K_{vco_example} := 10 \cdot \frac{MHz}{volt}$$

$d := \sqrt{2} \cdot A_V \cdot K_{vco_example}$ this is the calculated peak deviation of the FM signal

$d = 14.14214\,Hz$

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$f_m := 10 \cdot kHz$ this is the modulating/message frequency, the frequency offset at which a sideband will appear on the modulated carrier

$m \equiv \phi_d$ this is the modulation index, which is also equal to the phase deviation in radians

$m := \frac{d}{f_m}$ $m = 1.41421 \times 10^{-3}\,rad$ $m = 0.08103\,deg$

The sideband levels are simply equal to half of the modulation index.
For m small, <<1 , the higher higher order bessel functions are negligible, showing that the majority of the power is contained in the first sidebands. Thus, the higher order sidebands are considered insignificant

$sideband_level := 20\log\left(\frac{m}{2}\right)$

$sideband_level = -63.0103\,dBc$

The next equations compute the modulation index versus frequency for each resistor's & op-amp noise, The numerator equals the peak deviation.

$m_{R2}(f) := \frac{\sqrt{2} \cdot V_{noise_R2}(f) \cdot K_{vco}}{f}$ $m_{R3}(f) := \frac{\sqrt{2} \cdot V_{noise_R3}(f) \cdot K_{vco}}{f}$

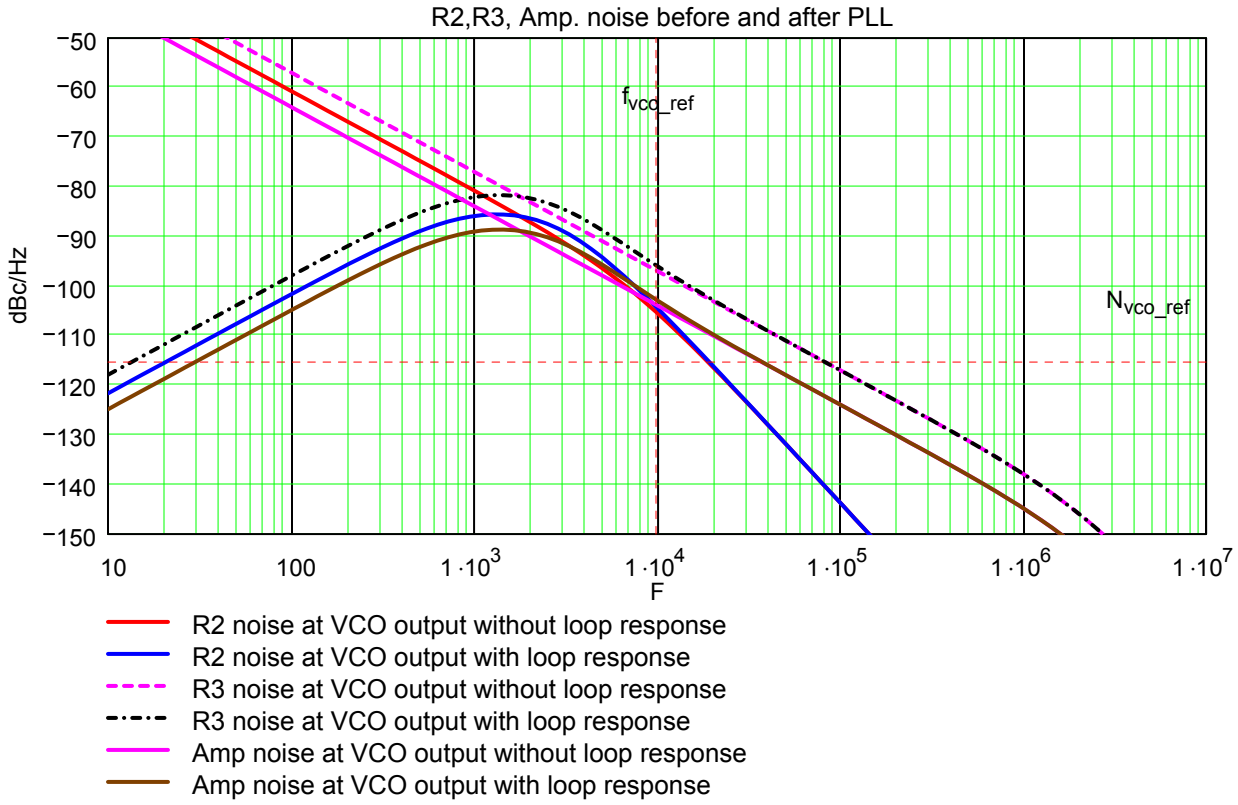
$m_{amp}(f) := \frac{\sqrt{2} \cdot V_{noise_amp}(f) \cdot K_{vco}}{f}$

The next three equations represent the SSB phase noise. They're the mod index divided by two, then multiplied by the transfer function of the PLL to this noise, exactly the same way as the PLL transfer function is applied to the inherent VCO noise. Within the loop bandwidth, the loop has a great deal of gain so the closed loop contribution (within the loop) of the resistor noise is greatly reduced. Outside of the loop, as $G_{ol}(f)$ loses steam, the resistor noise increases to the open loop value as if the PLL weren't even there.

$N_{R2}(f) := \frac{m_{R2}(f)}{2} \cdot \left|\frac{1}{1 + G_{ol}(f)}\right|$ $N_{R3}(f) := \frac{m_{R3}(f)}{2} \cdot \left|\frac{1}{1 + G_{ol}(f)}\right|$

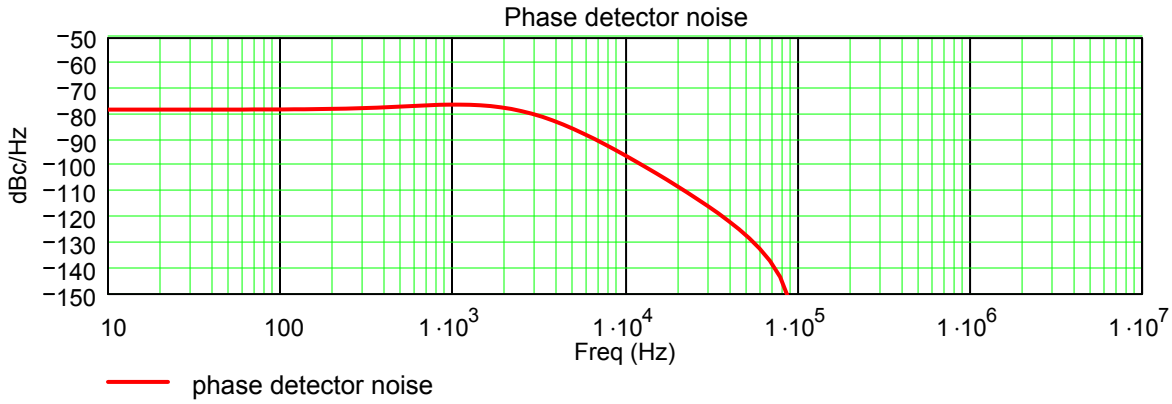
$N_{amp}(f) := \frac{m_{amp}(f)}{2} \cdot \left|\frac{1}{1 + G_{ol}(f)}\right|$

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phase detector contribution is flat across frequency, and is shaped by the PLL's closed loop response. According to national, the phase detector degrades at a 10*log(F_{ref}) rate not 20 log. This equation reflects that degradation versus frequency.

$$N_{p_d}(f) := 10^{\frac{N_{pd_ref}+10\cdot\log\left(\frac{f_{ref}}{Hz}\right)}{20}}\cdot G_{cl}(f)\cdot N$$



Optimum PLL loop bandwidth for lowest overall noise (estimate/starting point)

-Loop bandwidth is chosen as the frequency where the VCO noise and the open-loop PD noise cross over

$$BW_{opt} := \text{root}\left(\left|20\cdot\log\left(\left|N_{vco_alone}(x)\right|\right)\right| - \left|20\cdot\log\left(\left|N_{p_d}(0.1\cdot Hz)\right|\right)\right|, x\right)$$

$BW_{opt} = 141.78817\,Hz$

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This second estimate of optimum bandwidth includes the effect of resistor & amp noise when making the determination.....the problem with this of course is that it's always relative. The resistors will change with loop bandwidth, so this is only something that says "if you could narrow/widen the loop bandwidth with the same resistors, that might help" -- i.e. if you could change charge pump current for instance. Just something I threw in to see what it does with different designs. Realize that these approximations don't account for the phase detector noise rising near the loop BW either.

$$N_{vco_plus_Rnoise}(f) := \sqrt{\left(\left|N_{vco}(f)\right|\right)^2 + \left[\left(\left|N_{R3}(f)\right|\right)^2 + \left(\left|N_{R2}(f)\right|\right)^2 + \left(\left|N_{amp}(f)\right|\right)^2\right]}$$

$$BW_{opt2} := \text{root}\left[\left(\left|20\cdot\log\left(\left|N_{vco_plus_Rnoise}(x)\right|\right)\right| - \left|20\cdot\log\left(\left|N_{p_d}(0.1\cdot Hz)\right|\right)\right|\right), x\right]$$

sometimes this doesn't converge, I'm not sure why...

$BW_{opt2} =$ ▮ Hz

Now calculate the composite noise by adding the noise powers of the individual sources

$$N_{tot_no_R}(f) := mult\cdot\sqrt{\left(\left|N_{vco}(f)\right|\right)^2 + \left(\left|N_{tcxo}(f)\right|\right)^2 + \left(\left|N_{p_d}(f)\right|\right)^2}$$

$$N_{total}(f) := mult\cdot\sqrt{\begin{aligned}&\left(\left|N_{vco}(f)\right|\right)^2 + \left(\left|N_{tcxo}(f)\right|\right)^2 + \left(\left|N_{p_d}(f)\right|\right)^2 \ldots\\ &+\left[\left(\left|N_{R3}(f)\right|\right)^2 + \left(\left|N_{R2}(f)\right|\right)^2\right]\cdot Rnoise_enabled \ldots\\ &+\left[\left(\left|N_{amp}(f)\right|\right)^2\cdot loop_amp_enabled\right]\end{aligned}}$$

Phase noise at a few significant offsets

$20\cdot\log\left(\left|N_{total}(0.1\cdot kHz)\right|\right) = -77.88482$

$20\cdot\log\left(\left|N_{total}(1.0\cdot kHz)\right|\right) = -74.75507$

$20\cdot\log\left(\left|N_{total}(10\cdot kHz)\right|\right) = -92.60711$

$20\cdot\log\left(\left|N_{total}(100kHz)\right|\right) = -116.93043$

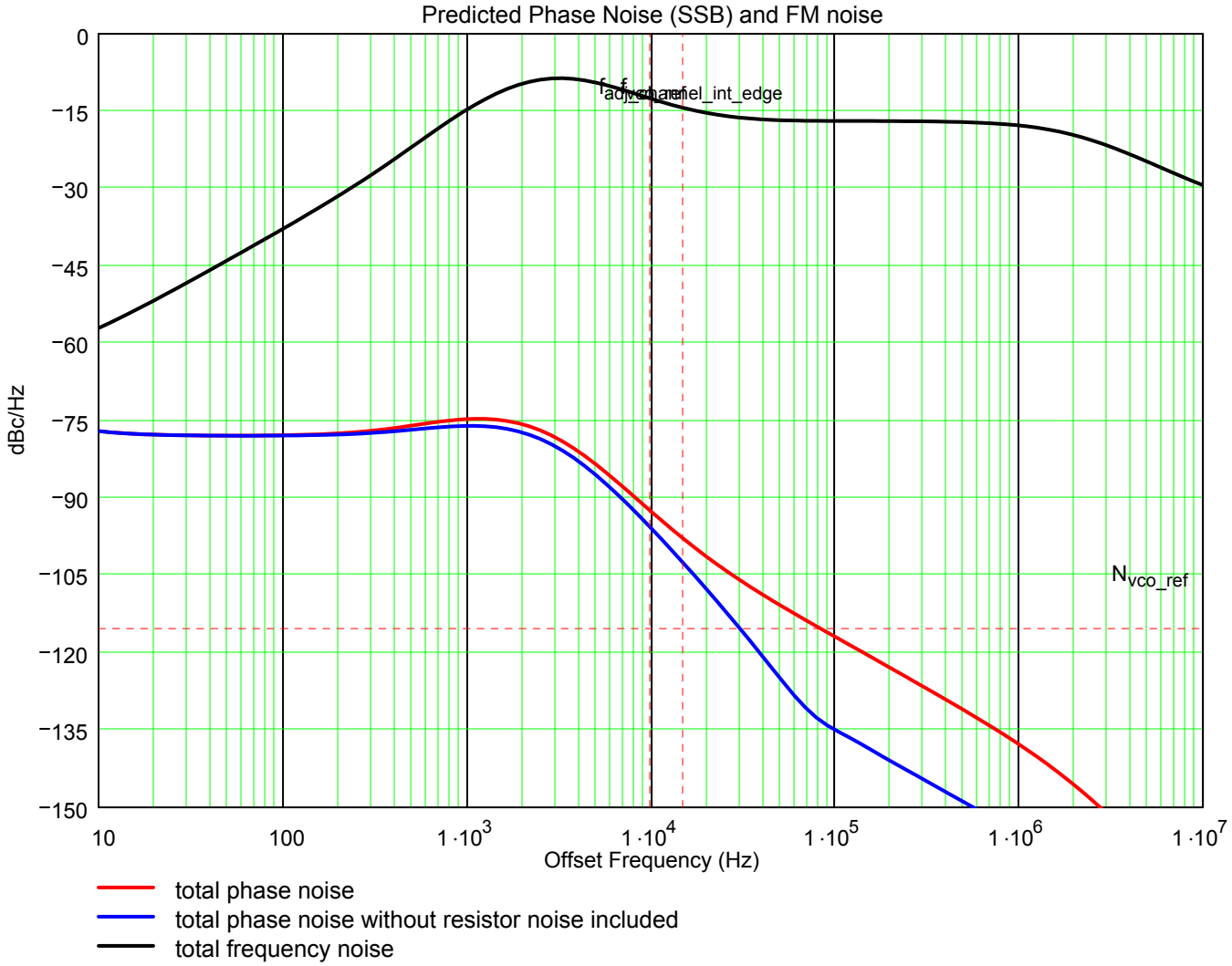
$20\cdot\log\left(\left|N_{tot_no_R}(0.1\cdot kHz)\right|\right) = -77.94557$

$20\cdot\log\left(\left|N_{tot_no_R}(1\cdot kHz)\right|\right) = -76.0628$

$20\cdot\log\left(\left|N_{tot_no_R}(10\cdot kHz)\right|\right) = -95.84235$

$20\cdot\log\left(\left|N_{tot_no_R}(100kHz)\right|\right) = -135$

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Now create a vector containing the phase noise data for this PLL so that it can be written to a file. This file can then be read in later, combined with the noise of another PLL, and the phase noise "metrics" can be calculated for the combined system. **Comment out the file write if desired.**

Note:

1) the frequency sweep parameters defined earlier in this file

2) The files written by the analysis worksheet (just below this) are written after the multiplier/divider has been accounted for

3) if you don't want to write the files after each analysis, "comment out" the file write commands using "disable evaluation"

Specify file to write here (this is from the analysis above. We then read the same file back in and use it to combine with a file written from a separate worksheet like this, perhaps an IF synthesizer)

this_synth := "synth_a" <-----File base name

write log sweep phase noise

$N_{new_total_vector_{spts}} := N_{total}\left(swp_{frq_{spts}}\right)$

write log sweep frequencies

$WRITEPRN(concat(this_synth, ".frq")) := \frac{swp_{frq}}{Hz}$

$WRITEPRN(concat(this_synth, ".pnd")) := 20 \cdot \log\left(N_{new_total_vector}\right)$

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SNR estimation based on integrated noise power *(for linear modulation schemes, not suitable for FM/FSK)*

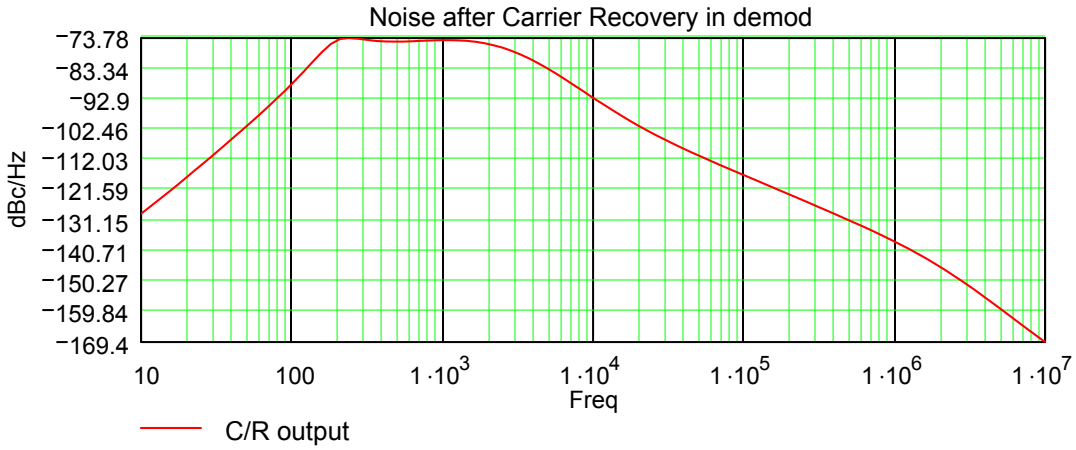
This now includes a 2nd order model for the carrier recovery PLL, so the interval of integration will be the max offset specification

Now add in the transfer function of a 2nd order highpass filter (the equivalent transfer function for phase noise attenuated by the carrier recovery loop in a QAM/QPSK system)

$$H_{cr}(f) := \frac{\left(\frac{2\cdot\pi\cdot f\cdot i}{f_{cr_bw}\cdot 2\cdot\pi}\right)^2}{1+\left(\frac{2\cdot\pi\cdot f\cdot i}{f_{cr_bw}\cdot 2\cdot\pi}\right)\cdot\zeta_{cr}+\left(\frac{2\cdot\pi\cdot f\cdot i}{f_{cr_bw}\cdot 2\cdot\pi}\right)^2}$$

$$N_{CR_out}(f) := N_{total}(f)\cdot\left|H_{cr}(f)\right|$$

This is the equivalent noise seen at the demodulator output (the carrier recovery loop acts as a highpass filter for the phase noise)



Integrated noise calcs will be both for a variable upper limit of integration and a variable lower limit of integration. To see the real effects of the lower limit of integration, set the carrier recovery loop bandwidth to be low so that highpass doesn't conflict with your study... (that's set at the top of the worksheet in the user input area)

$$N_{integrated}(f_{ll}, f_{ul}) := \frac{1}{Hz}\cdot\int_{f_{ll}}^{f_{ul}}\left(\left|N_{CR_out}(f)\right|\right)^2 df$$

$$N_{integrated}(f_{start}, f_{stop}) = 1.11246\times 10^{-4}$$

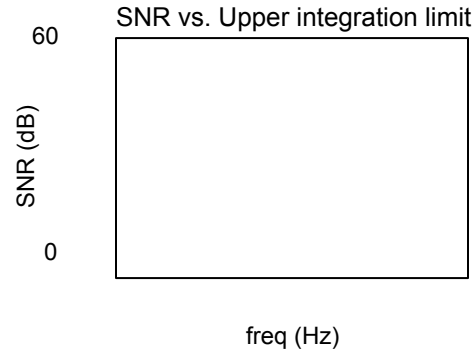
the -3 dB is because for SNR there is really twice as much noise than the SSB noise

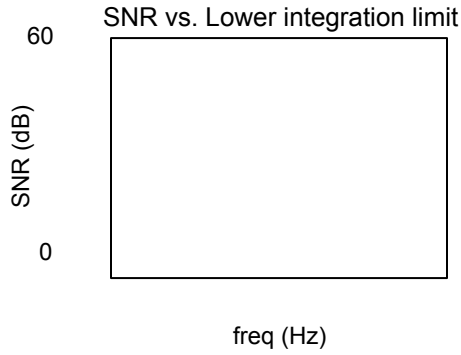
$$SNR_est(f_{begin}, f_{end}) := 10\cdot\log\left(\frac{1}{N_{integrated}(f_{begin}, f_{end})}\right) - 3$$

$$SNR_est(f_{start}, f_{stop}) = 36.53716$$

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Now look at how the SNR is affected by the upper limit of integration (if the curve of SNR v.s. upper integration limit ever starts rising again, set the TOL variable at the top of this worksheet. This controls how mathcad does the numerical integration) (these graphs have been disabled, re-enable if you want)





RMS phase error integrated over the frequencies specified at the beginning of this worksheet will be calculated. Here's my explanation of the equation.

1. factor of two in front of the Ntotal(f)^2 is to double the SSB phase noise to get both SSB's contribution..
2. the Ntotal^2 term is to convert noise voltage to power (since all of these equations refer to PSD)
3. Basically all the noise is summed and the RMS value is taken to be the "RMS" phase error, simple as that. It's always scary boiling down a complex curve to a single number.

$$\sigma_\phi := \frac{1}{\sqrt{Hz}} \cdot \sqrt{2 \left[\int_{0 \cdot Hz}^{f_{stop}} \left(\left| N_{CR_out}(f) \right| \right)^2 df \right]}$$

RMS phase error

$\sigma_\phi = 0.85563\,deg$

Here s another approximation of SNR I found in William Reuters paper, apparently taken from a Feher reference

$$SNR_{estimate_2} := 20 \cdot \log\left(\frac{1}{\sigma_\phi \cdot rad} \right)$$

$SNR_{estimate_2} = 36.51672$

Residual FM (RMS deviation)

This one is identical to the RMS phase error calculation except for the following

1. f^2 term in the integral -- accounts for how fm deviation vs. offset from the carrier
2. Lower limit of integration is 1 Hz (change limits to suit your system)
3. no CR loop is applied since this is not applied to coherent systems

$$\sigma_f := \frac{1}{\sqrt{Hz}} \cdot \sqrt{2 \cdot \int_{1 \cdot Hz}^{f_{stop}} \left[(f)^2 \cdot \left(\left| N_{total}(f) \right| \right)^2 \right] df}$$

$\sigma_f = 349.11805\,Hz$ RMS deviation

now caluclate SNR based on this for a FM/FSK system

$$SNR_{FM} := 20 \cdot \log\left(\frac{deviation_{pk} \cdot \frac{\sqrt{2}}{2}}{\sigma_f} \right)$$

$SNR_{FM} = 15.67268$ SNR in dB

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Adjacent channel rejection calculations..

(calculate an approximate adjacent channel rejection due to reciprocal mixing... something that happens before IF filters can help you at all). Very important on HF where the channels are close!
This is something I just threw together, I think it's representative of the amount of noise you would see on channel if a carrier were placed on the adjacent channel....to figure out the actual rejection with a real system, fill in the cochannel variable at the beginning of the worksheet.

$$p_{adj_chan} := \frac{1}{Hz} \cdot \int_{f_{adj_channel_int_edge}}^{f_{adj_channel_int_edge}+f_{adj_ch_bw}} \left(\left| N_{total}(f) \right| \right)^2 df$$

$p_{adj_chan} = 1.08832 \times 10^{-6}$

Rejection in dB

$$Rejection_{adj_chan} := 10 \cdot \log\left(\frac{1}{p_{adj_chan}} \right) + Rejection_{co_channel}$$

$Rejection_{adj_chan} = 47.63244$

Two Point Modulation (Lance's best attempt.) Note the R-C in series with the TCXO path, that's so you can delay/filter one path relative to the other and see what it does to the frequency response...

constants for scaling modulation to VCTCXO and vco tuning

$k_{vco_mod} := 1$ $k_{tcxo_mod} := 1$

assume an the tcxo has a frequency response that can be modeled as a simple RC filter

$R_{tcxo} := 1 \cdot k\Omega$ $C_{tcxo} := 0.01 \mu F$

$$G_{tcxo_rc}(f) := \frac{1}{(2 \cdot \pi \cdot f \cdot i \cdot R_{tcxo} \cdot C_{tcxo}) + 1}$$

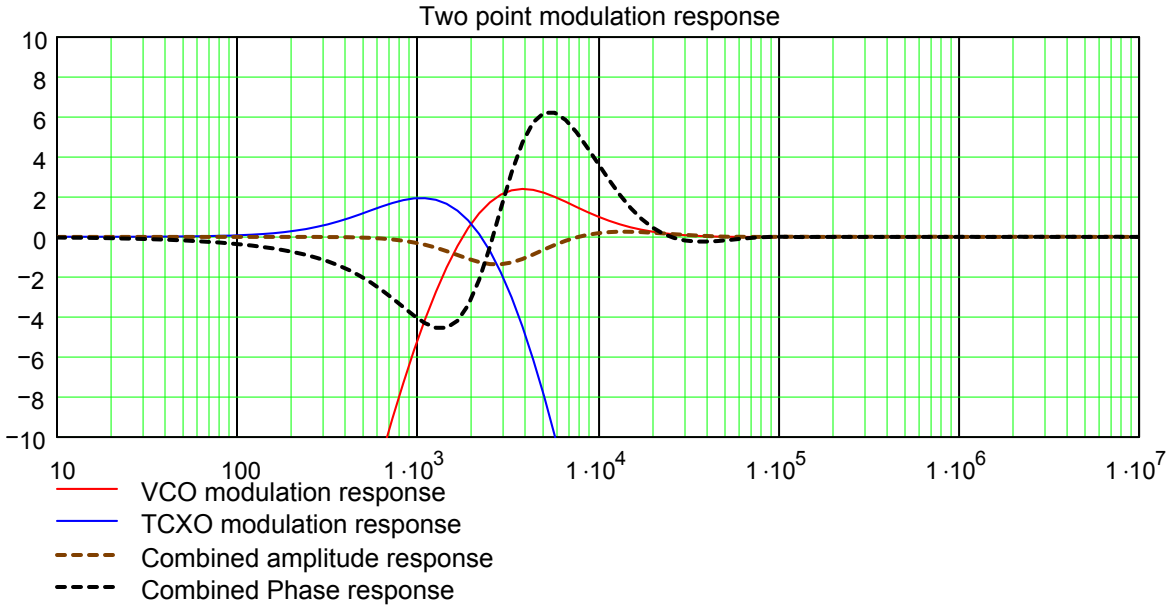
$$f_{c_tcxo} := \frac{1}{2 \cdot \pi \cdot R_{tcxo} \cdot C_{tcxo}}$$

$f_{c_tcxo} = 15.91549\,kHz$

$$modulation_{vco}(f) := k_{vco_mod} \cdot \frac{1}{1 + G_{ol}(f)}$$

$$modulation_{tcxo}(f) := k_{tcxo_mod} \cdot G_{cl}(f) \cdot G_{tcxo_rc}(f)$$

$$modulation_out(f) := modulation_{vco}(f) + modulation_{tcxo}(f)$$



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Note: The transient analysis below *can* lead you to false conclusions at times. The assumption that must be valid for this analysis to be "decent" is that there is no saturation occurring. See www.radiolab.com.au for a good application note on this, and software that can handle these conditions as well. ***Note: the analysis here is probably not accurate for the "amplifier" case, so beware.***

Transient Analysis stolen from Dean Banerjee's PLL book (see references) Only minor changes were made to fix some bugs (perhaps they were mathcad 8 vs. mathcad 7 incompatibilities?)
I also changed the names of some variables to make it compatible with my worksheet
LL 8/1/1999

User Enters these

$f2 := f_{out}$ Final Frequency

$f1 := f_{out} + 5\cdot MHz$ Starting Frequency

$t_pts := 12500$ number of points in time domain

$t_max := 15\cdot msec$ maximum time displayed

$ftol := 1000\cdot Hz$ Tolerance Marker for Lock Time Measurments

Calculations

$R3 := \max\left(\begin{pmatrix} R_3 \\ 1\cdot\Omega \end{pmatrix}\right)$ $C3 := \max\left(\begin{pmatrix} C_3 \\ 1\cdot pF \end{pmatrix}\right)$

$den2 := C_1 + C_2 + C3$

$N := \frac{f2}{f_{ref}}$ $den4 := R_2\cdot R3\cdot C_1\cdot C_2\cdot C3$

$den3 := C_2\cdot C3\cdot R_2 + C_1\cdot C_2\cdot R_2 + C_1\cdot C3\cdot R3 + C_2\cdot C3\cdot R3$

$den1 := \frac{K_\phi\cdot K_{vco}\cdot C_2\cdot R_2}{N}$ $num0 := \frac{K_\phi\cdot K_{vco}\cdot(f2 - f1)}{N}$

$den0 := \frac{K_\phi\cdot K_{vco}}{N}$ $num1 := num0\cdot R_2\cdot C_2$

$$v := \begin{pmatrix} \frac{den0}{den4}\cdot sec^4 \\ \frac{den1}{den4}\cdot sec^3 \\ \frac{den2}{den4}\cdot sec^2 \\ \frac{den3}{den4}\cdot sec \\ 1 \end{pmatrix} \qquad v = \begin{pmatrix} 2.57666\times 10^{19} \\ 7.32325\times 10^{15} \\ 5.85119\times 10^{11} \\ 1.31518\times 10^{7} \\ 1 \end{pmatrix}$$

$p := polyroots(v)\cdot sec^{-1}$

These are the poles

$$p = \begin{pmatrix} -1.31072\times 10^{7} \\ -2.60581\times 10^{4} \\ -1.25101\times 10^{4} \\ -6.03038\times 10^{3} \end{pmatrix} time^{-1}$$

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$$A_0 := \frac{\frac{num0}{den4}}{(p_0 - p_1)\cdot(p_0 - p_2)\cdot(p_0 - p_3)} \qquad A_0 = 5.74092 \times 10^{4}\,time^{-2}$$

$$A_1 := \frac{\frac{num0}{den4}}{(p_1 - p_0)\cdot(p_1 - p_2)\cdot(p_1 - p_3)} \qquad A_1 = -3.62978 \times 10^{10}\,time^{-2}$$

$$A_2 := \frac{\frac{num0}{den4}}{(p_2 - p_0)\cdot(p_2 - p_1)\cdot(p_2 - p_3)} \qquad A_2 = 1.12073 \times 10^{11}\,time^{-2}$$

$$A_3 := \frac{\frac{num0}{den4}}{(p_3 - p_0)\cdot(p_3 - p_1)\cdot(p_3 - p_2)} \qquad A_3 = -7.57757 \times 10^{10}\,time^{-2}$$

4 Pole Analysis

$k := 0 .. t_pts$

$t_k := k \cdot \frac{t_max}{t_pts}$

$i := 0 .. 3$

$$F(t) := f2 + \sum_i A_i \cdot e^{p_i \cdot t}\left(\frac{1}{p_i} + R_2 \cdot C_2\right)$$

$span1 := 10 \cdot MHz$

Vertical Span of the Plot

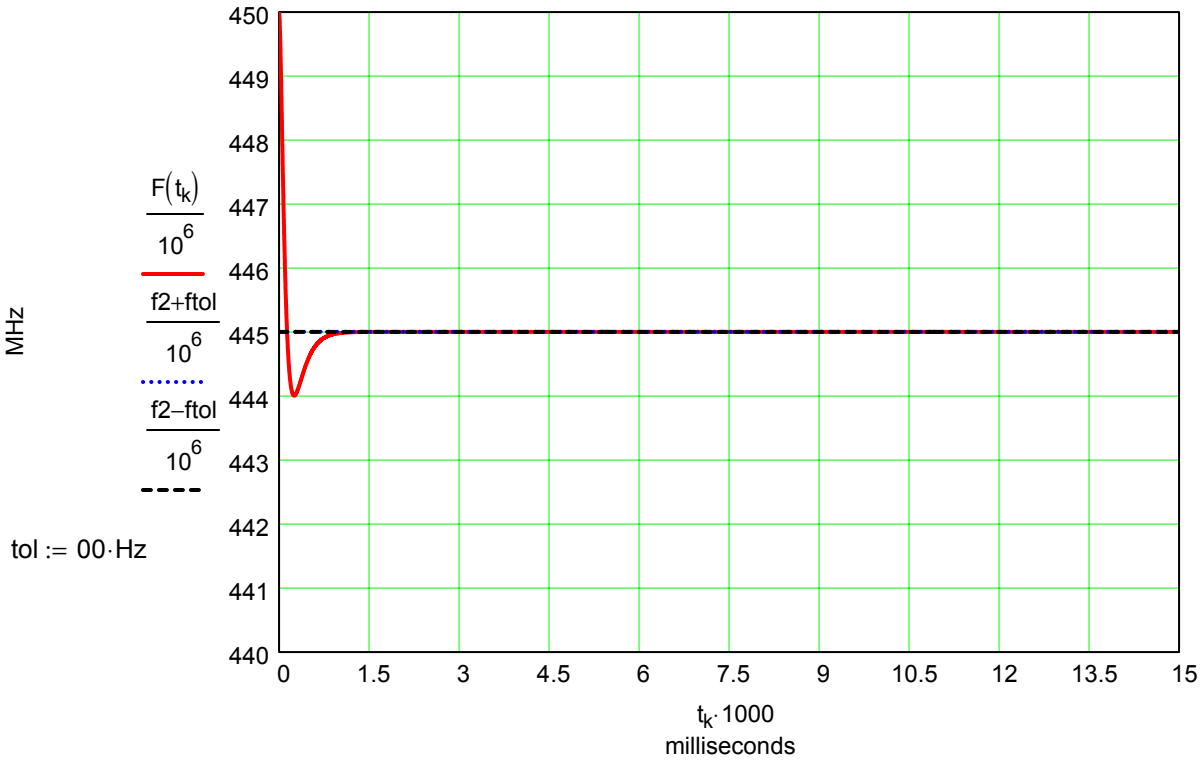
$center1 := f2$

Center Frequency

For whatever reason, the traces sometimes disappear on these. Try adjusting the scale to see if the traces will reappear. I have found that this is a bug in mathcad and/or an interaction with various video settings. I suggest seeing the mathcad site for more info

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PLL Transient Response



tol := 00·Hz

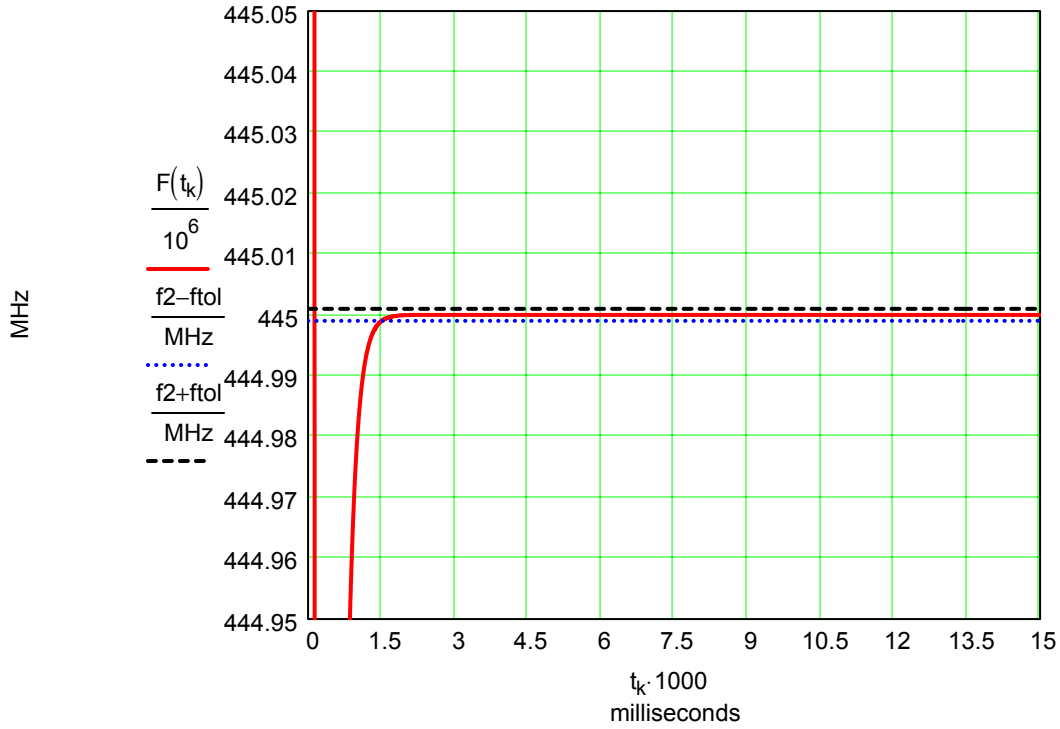
span2 := 100·kHz

center2 := f2

Vertical Span of the Plot

Center Frequency

PLL Transient Response



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